POLICY/LEGAL FRAMEWORK FOR SPACE TOURISM REGULATION

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I. INTRODUCTION

The concept of citizens being able to "tour the heavens" sometime not too far into the 21st century is one that is exciting to millions and may well entice tens of thousands to try. But whether the tourism is in an orbiting hotel, a reusable launch vehicle [RLV] or something more modest such as a ride in balloon above 100,000 feet, one thing is clear, there will be government regulation of these activities.

Why and to what extent regulation will be imposed will to a large degree depend on the policy and legal framework for such regulation and the degree and specificity of guidelines that flow from it. While marketeers and manufacturers are pursuing customers and hardware designs, a concomitant effort to ensure that Space Tourism projects do not inadvertently proceed into a regulatory box canyon is essential.

So, what do we mean by "policy/legal framework?" Basically, that there is a hierarchy of international agreements, policies, laws and regulations that control space activities of citizens. And why do we need this layering of red tape? It's because activities that involve public safety, economic activity, the environment, national security or foreign policy, all require a regulatory framework to assure operation that redounds to the public good. So where do we begin to look for this Policy/Legal Framework? An authoritative place to begin might be the 1967 Outer Space Treaty *[Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies].* For a long time it was the only governing document that covered commercial space activities. For example, when Space Services, Inc. of America decided to conduct the first commercial space launch in 1982, there were no statutes, policies, or regulations on the books anywhere. Only this Treaty, which holds that States are responsible for the activities of their own government, *or non-government* entities' activities. On its own, SSI undertook to contact and get clearance from all the US government agencies it thought might have concerns about a commercial launch. Agencies contacted included the State Department; North American Aerospace Defense Command; the US Navy;

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Bureau of Alcohol, Tobacco and Firearms; the Coast Guard and a number of others. The problem was that there were no standards, guidelines or anything upon which agencies could or should base a decision. In the end, in only five months, concurrence was received and Conestoga I was launched into sub-orbit out over the Gulf of Mexico.

As is well known, that launch prompted the Reagan Administration to eventually push for legislation establishing a central regulating authority in the Department of Transportation under the 1984 Commercial Space Launch Act. Since that time the Act has been amended and recodified in Title 49, Subtitle IX of the US Code. Now, however, we are facing a new challenge. Space Tourism is uncharted territory. Let's start with the Outer Space Treaty and see what new issues present themselves when tourists want to travel or vacation in space.

II. OUTER SPACE TREATY ISSUES.

States' Responsibility. We've already discussed the basic requirement of the Treaty that States [countries rather than the 50 US states] are responsible for their non-governmental entities' [NGE's] activities in space. It is not much of a stretch to assume that both passengers and crew of Space Tourism vehicles or facilities [orbiting habitats] would be covered by the "entities" provision. But does "being responsible" mean that, if it wanted, a country could chose a *laisse-faire* approach? Apparently, anticipating possible difficulties in this area, the Treaty does specify, "The activities of nongovernmental entities in outer space. . . shall require authorization and continuing supervision by the appropriate State Party..." [Article VI]

Liability. Here the Treaty provides that States are internationally liable for damage to another State or its natural or juridical persons by objects launched into outer space. The liability goes to the launching State or State from which the launch takes place. If the State is liable, there is a natural supposition that it would hold some liability leverage over its NGE's that engage in space activities. However, here it is unclear what liability might accrue in the event an NGE space mission resulted in harm to a foreign citizen. This suggests perhaps some sort of Warsaw Convention guidelines might be needed regarding, for instance, how much monetary compensation a person might be entitled to in the event of an international space touring accident.

Jurisdiction and Control. Unlike the Law of the Sea with, depending on the State, either a 12 mile limit or a 200 mile limit for jurisdiction, the Treaty [Article VIII] states that, "A State on whose registry an object launched into outer space is carried shall retain jurisdiction and control over such object, *and over any personnel* thereof, while in outer space.. ." So, would this mean that the US, for example, would retain jurisdiction and

control over all space tourists on US NGE missions, regardless of their nationality? What and who would have jurisdiction and control if an international passenger on a US Space Tourist mission committed a crime against another international tourist? In this instance, what legal sanctions would apply? National, state, both? These are policy/legal framework issues that must be addressed at some juncture in the future.

Assistance to Astronauts. This seems straightforward on the surface, yet could be one of those issues that needs consideration. Article V of the Treaty provides that States shall render to astronauts all possible assistance in the event of accident, distress or emergency landing on the territory of another State or high seas." Very straightforward; however, what is an astronaut? Webster's defines astronauts simply as," A person trained to make rocket flights into outer space." And that definition surely fits the Space Shuttle crew today, they are highly trained and the Shuttle is launched by solid rocket motors and onboard main engines that function like rockets. The key distinction is "highly trained." Space tourists are not going to be highly trained, although there would be some brief familiarization with emergency procedures as we have on airliners today. So, although logic would lead one to the conclusion that the provisions of this Article would apply to passengers as well as "astronauts", it is a lacuna in the policy/legal framework that will need to be filled.

Cooperation and Mutual Assistance. This Article gives a State the right to seek consultation with another State if activities planned would cause harmful interference in the peaceful use of outer space. Although it seems prosaic, this is a potentially important provision for Space Tourism. When a new commercial market is being developed, many creative ways to reap economic benefit emerge, often far beyond the scope of policymakers' imaginations. It is not unreasonable to expect some visionary entrepreneur to develop a project that might impinge on the sensibilities of another sovereign State. The Treaty ensures that State to State consultation can be arranged to address any troublesome issues.

III. NATIONAL SPACE POLICY ISSUES.

The National Space Policy of 1996 contains a number of provisions designed to promote and encourage commercial space activity. However, nowhere does it address the prospect of private citizens traveling into outer space on commercial vehicles. A brief review of some of the salient points actually in the Policy might give some indication of what additions or modifications will be needed for Space Tourism.

Access to and Use of Space. The Introduction states that access to and use of space is central for [among other things] preserving commercial interests. The US will pursue greater levels of partnership and cooperation

to ensure the continued use of outer space for peaceful purposes. What could be more peaceful use than Space Tourism? When this Policy was signed in 1996, no one contemplated Space Tourism. But, by the time that technology and the market have advanced sufficiently for Space Tourism missions to become reality, the National Space Policy of that era will need to include specific reference to the issue.

Goals of the US Space Program. Apart from the military and scientific aspects, current policy includes enhancing economic competitiveness, encouraging private sector investment in use of space technology and promoting international cooperation. These are all grand, broad-brush statements, but when hundreds, or even thousands of citizens are clamoring for a ride to the heavens, the topic will be important enough to mention specifically.

Regulation of Commercial Space Activities. Here the Policy has a reasonable approach. Regulation shall be supervised or regulated only to the extent required by law, national security, international obligations and public safety. With Space Tourism there will be two dimensions that have not yet been specifically addressed: safety of tourists, and economic activity of the carriers providing the touring services. The reasons for these omissions go back to the previously mentioned fact that this Policy did not contemplate private citizens traveling in space for pleasure. Also, the Policy calls for the government to identify and propose appropriate amendments to or elimination of US laws and regulations that unnecessarily impede space sector activities. This is an important provision that should remain. Whereas, since the lives and safety of ordinary people are going to be involved in these missions, there will be a natural inclination to impose a regulatory regime to preserve their safety to the utmost possible. However, it is important that, while assuring safety to a reasonable, but not absolute degree [as with aviation], regulations should not become so onerous and burdensome that a budding industry is smothered before it has a chance to flower. And if a very conservative regulatory regime is later recognized as being unnecessarily burdensome, it is incumbent upon government agencies to propose amendments to or elimination of regulations and laws that unnecessarily impede space sector activities.

US Government Fosters International Competitiveness. While recognizing that the Government will not directly subsidize US Space Tourism, there are things it can do to foster competitiveness. When new facilities or launch vehicles are being developed by the government, there often will be commercial considerations that, if factored into government decisions, can materially benefit the Space Tourism industry.

IV. LEGISLATIVE ISSUES AFFECTING SPACE TOURISM.

Treaties and National Space Policies set the general direction for allowing space activities and establishing appropriate general constraints to preserve public safety and well being. However, the actual regulation of citizen's activities comes from statutory authority given to a responsible government agency. Unlike 1982, when SSI was launching the Conestoga rocket, today there is a designated responsible agency, and it has authorizing legislation. Title 49, Subtitle IX of the US Code provides the Secretary of Transportation [DOT] and his executive agent, the Administrator of the Federal Aviation Administration [FAA], with authority and responsibility to support and regulate commercial space launch activity. Space Tourism will be a subset of launch activity, and it is difficult to imagine that any agency but DOT/FAA will have the mandate to oversee it.

Looking broadly at the current legislation, not surprisingly, there are a number of provisions that reflect the concepts and ideas set forth in the National Space Policy that will easily apply to Space Tourism. However, since commercial space transportation presently does not involve launching humans into space, some new provisions will be required to implement provisions of the Policy.

Some of the issues and concepts that Congress had in mind when it enacted the legislation can briefly be summarized.

Peaceful uses of outer space are beneficial. Those uses and the private application of technology have achieved economic benefit.

New technologies and services are being developed for traditional space activities. However, the specific inclusion of commercial activities involving humans in space, to include Space Tourism, is needed. It is reasonable that the primary focus of new legal language should be "commercial activities involving humans," of which Space Tourism is a major, but not exclusive element. By the time that Space Tourism missions begin, there may already be commercial activity in space, perhaps in the International Space Station.

New launch vehicles and services will enable the US to retain its international competitive position. Space Tourism would certainly fall under the "new services" category and needs mentioning.

The US should encourage private sector launches and associated services [Space Tourism] and only regulate them to insure compliance with international agreements and preserve public health and safety. This virtually mirrors the National Space Policy.

Space transportation is an important element of the national transportation system. Certainly the RLVs that will be used for Space Tourism will be a very important element, and will open the interior of the

US to space activity and opportunities in ways that were not possible with the use of expendable launch vehicles [ELV's]

State governments have the ability to encourage and support private sector space activities. Space Tourism will be another such activity.

What authority has Congress given the Secretary of Transportation specifically? Basically, to encourage, facilitate and promote commercial space launches. He is to take actions to facilitate private sector involvement in commercial space activity and promote public-private partnerships with the US Government, state governments, and the private sector to build, modernize and expand or operate a space launch infrastructure. Regulatory authority stems from the legal requirement for any US citizen or anyone in the United States or its territory desiring to conduct a commercial launch or operate a launch site to obtain a license. US citizens can conduct launches or operate launch sites outside the US if an agreement has been reached with the foreign government that it has jurisdiction over such activities. Space Tourism missions will have to have launch licenses. Another requirement is that payloads must comply with US laws. This is a new area with potential Space Tourism implications that will also have to be addressed [i.e. items, (payload) that passengers would be prohibited from bringing aboard].

An important authority is the ability to stop any launch if it appears unsafe. With commercial human space travel, and particularly Space Tourism, safety will be paramount. Man-rated missions will come under critical scrutiny, as the safety requirements will be much higher. A 95% launch success rate is not nearly enough for operations involving humans.

V. REGULATORY ISSUES.

The final level of the policy framework encompasses the actual regulations themselves and the issues they must address. The Treaty laid certain responsibilities on the nation, the Policy outlined goals, the Statute specified responsibilities and authority. But for all this to be meaningful there must be specific regulations. And, deriving from the Statute, these regulations will have to address all the important issues and concerns involved with transporting humans into outer space for pleasure. So, what are some of those issues?

Passenger safety will be the primary issue. No matter what the mode of Space Tourism, reusable launch vehicle, balloon, or aircraft, safety of the passenger will be the paramount regulatory issue that the industry must address. People are allowed to take virtually unlimited risks on their own. But a commercial entity that carries passengers must demonstrate some threshold level of assurance that the people who pay for the trip have a reasonable chance of completing it safely. For Space Tourism we have

several areas of safety to consider. First is safety of passengers, second crew and lastly third parties.

Safety of passengers will rest on three immediate considerations: reliability of the RLV; training and skill of the flight and ground crews; and, adequacy of the launch and landing sites. In addition, although there is no analogous requirement for airline flight, there may also be some minimum physical and psychological requirements for passengers. Even as tourists, passengers may be subject to stresses that they have never experienced before. Space travel is unlike any previous form of tourism. If passengers were denied tickets on the basis of not meeting some physical/psychological standard for space flight, there might even be lawsuits under, say, the Americans With Disabilities Act. Some disabilities might be manageable on a Space Tourism mission, others might not. Another issue could be whether children would be allowed on Space Tourism missions, and if so, would there be a minimum age? Would a passenger who could otherwise afford it be allowed to bring a pet on board? These are issues that the airlines have had to deal with, and it's quite possible that Space Tourism missions could involve the same kinds of concerns. In any event, the regulatory process by which the determination is made regarding physical/psychological requirements for crew and passengers will undoubtedly entail a great deal of effort. Nevertheless, this is an area where it will be important to get the right balance between a regulatory regime that protects passengers and crew and yet is not so burdensome that the missions become economically infeasible. And, operating guidelines will have to ensure a reasonable degree of safety for uninvolved third parties, as is the case with the airline industry today. These missions will be flying over populated areas; perhaps even using major airports as launch and landing sites. Third parties will be at some potential risk.

Certification Issues. All common carriers have to meet certain standards of safety before they are certified to carry commercial passengers. Airliners are the most obvious example, and there are rigorous standards that all new aircraft designs have to meet before they can be sold to airlines for passenger service. The same thing will be true of RLV's that will carry tourists. There are already some regulations in place for commercial balloon operations; although, it is not apparent whether these cover commercial operations at extremely high altitudes with outer space-like environment. In any event, a host of issues arises when one contemplates government certification of an RLV for commercial passenger service. Since the FAA regulates commercial space launches, it is not unreasonable that their mandate would likely be expanded to certify launch vehicles as well as license launch operations and sites.

Certification of the Launch Vehicle. In this arena, a number of immediate questions arise. How many flights of an RLV would be required

to certify its spaceworthiness? Who would develop the methodologies for analyzing safety and engineering issues? The FAA? The manufacturer? Expendable launch vehicles always have a destructive flight termination system [FTS]. Obviously, a non-destructive FTS would be needed for an RLV. What kinds of these non-destructive systems could be devised? What sort of Safety System Process would be needed to define the safe operating envelope of the RLV? Recognizing that no transportation system is 100% safe, some acceptance of potential accident is necessary. How safe would an RLV Space Tourism mission have to be? How remote would the likelihood of death or injury have to be? One in a million missions; one in 10 million? What would be an acceptable level of casualties per million flights? And, how would it be possible to determine when an RLV system had reached the acceptable level of safety and reliability?

Launch Sites. Launch site selection and operation will depend on a complex mix of technological, economic, environmental and legal factors. RLV's are more efficient at high altitude launch sites because every foot above sea level is a foot that the RLV engines do not have to travel to get into space. The same vehicle could transport a greater load into orbit from Denver than Cape Canaveral because it had a vertical mile less to go. Depending on the launch technology, operating from a major airport may be possible, but deconflicting it with airline traffic would be required. An assessment of the environmental impact of operating at a major airport may or may not be favorable for RLV operations. Special launch/recovery sites at more remote locations may be necessary. Economically, the sense of adventure for a Space Tourist might be greater at a remote "spaceport" than at a major airport. And there could be greater operating fees at a major airport than out on the [space] range. State and local tax laws and other restrictions may also affect where a Space Tourism company decides to operate. These are normal market forces that will accommodate the policy framework in a manner that is best for themselves.

Environmental Impact. All human activity has real or potential effect on the environment. The Outer Space Treaty reflects some environmental concerns. States should avoid bringing harmful contamination back from outer space that might adversely affect the environment. However, launch and landing activities will be subject to Environmental Law, as is already the case with ELV launches and site operations. As the scope and details of proposed Space Tourism projects emerge, it will be incumbent upon the operators to consult with the FAA and other government offices to make certain that environmental concerns have been satisfied. Indeed, satisfactory environmental impact studies will most certainly be a *sine qua non* for obtaining a launch license. And, apart from any government objections, projects can also be thwarted in the courts by other concerned citizens who may not appreciate the benefits of Space Tourism operations.

International Considerations. The Outer Space Treaty covers many salient aspects of activity, but there are others not contemplated by the Treaty authors that affect Space Tourism. For example, if a Space Tourism firm were offering a landing in a different country than the launch, would there be customs and immigrations issues to be worked out? That is, would passengers need passports? Would their belongings be subject to customs duties at the destination? Again, these are new kinds of issues that always emerge when a new industry is born.

How Much Liability? As the commercial launch industry was developing, a major hurdle was the potential for exposure to third party liability. The favorite scenario was a failed rocket launch that fell from Cape Canaveral back into the Miami Convention Center where the American Bar Association was holding its annual meeting. Thus, Title 49 contained a valuable liability safety net that industry regarded as indispensable for operations in its early days. As is generally known, the FAA Commercial Space Transportation Office after analyzing the license application, determines the amount of third party liability a company must secure in order to obtain a launch license. Once the insurance coverage is obtained, if there were a catastrophic accident and third party liability exceeded the insurance coverage specified in the license, the government, subject to appropriation, would provide indemnity up to $1.5 billion above the insurance level specified in the license. There has never been a catastrophic failure requiring significant third party liability claims to be paid. So, an important question is whether the Space Tourism industry will feel the need for similar indemnification protection before it can undertake operations. The potential for liability would seem perhaps higher, given that passengers will be involved. The Outer Space Treaty specifies that States are liable for the space activities of their entities, so some legislation covering the possibility of an accident on foreign territory would seem to be required. A separate convention on Space Tourism liability may need to be convened, the Warsaw Convention might be amended to cover space tourism, or some other international agreement may be reached on liability in the event a passenger were killed or injured on a mission. Perhaps an amendment to Title 49 would suffice. However, the issue of liability for accidents in the United States will likely be the subject of great debate in the Administration and the Congress.

VI. SUMMARY.

While most of the attention and effort will be focused on the traditional issues of technology, product development, market research and financial backing, for Space Tourism to be successful there must be an adequate policy/legal framework within which these activities can take place and succeed. Beginning with our international obligations under the